PRACTICAL NUMBER: 9

AIM: Implement following application using suitable data structures

i. Online shopping

ii. Theater Ticket booking

iii. Cab booking application

iv. Railway reservation system

PROGRAM*: (*Railway reservation system)

```
//SHEIKH WASIMUDDIN 23071044 60 CSEIOT//
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
struct Train {
    int trainNumber;
    char trainName[50];
    int seatsAvailable;
    struct Train* next;
};
struct Train* createTrain(int trainNumber, const char* trainName, int
seatsAvailable);
void addTrain(struct Train** head, int trainNumber, const char*
trainName, int seatsAvailable);
```

```
void bookTicket(struct Train* head, int trainNumber);
void displayTrains(struct Train* head);
void freeTrains(struct Train* head);
int main() {
    struct Train* head = NULL;
    int choice, trainNumber, seatsAvailable;
    char trainName[50];
    do {
        printf("\nRailway Reservation System\n");
        printf("1. Add Train\n");
        printf("2. Book Ticket\n");
        printf("3. Display Available Trains\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter Train Number: ");
                scanf("%d", &trainNumber);
                printf("Enter Train Name: ");
                scanf("%s", trainName);
                printf("Enter Seats Available: ");
                scanf("%d", &seatsAvailable);
                addTrain(&head, trainNumber, trainName, seatsAvailable);
```

```
            printf("Train added successfully!\n");
            break;
        case 2:
            printf("Enter Train Number to book: ");
            scanf("%d", &trainNumber);
            bookTicket(head, trainNumber);
            break;
        case 3:
            printf("Available Trains:\n");
            displayTrains(head);
            break;
        case 4:
            printf("Exiting the system.\n");
            break;
        default:
            printf("Invalid choice! Please try again.\n");
    }
  } while (choice != 4);
  freeTrains(head);
  return 0;
}
struct Train* createTrain(int trainNumber, const char* trainName, int
seatsAvailable) {
  struct Train* newTrain = (struct Train*)malloc(sizeof(struct Train));
```

```
    newTrain->trainNumber = trainNumber;
    strcpy(newTrain->trainName, trainName);
    newTrain->seatsAvailable = seatsAvailable;
    newTrain->next = NULL;
    return newTrain;
}
void addTrain(struct Train** head, int trainNumber, const char*
trainName, int seatsAvailable) {
    struct Train* newTrain = createTrain(trainNumber, trainName,
seatsAvailable);
    newTrain->next = *head;
    *head = newTrain;
}
void bookTicket(struct Train* head, int trainNumber) {
    struct Train* current = head;
    while (current != NULL) {
        if (current->trainNumber == trainNumber) {
            if (current->seatsAvailable > 0) {
                current->seatsAvailable--;
                printf("Ticket booked on %s (Train No: %d). Seats remaining:
%d\n", current->trainName, trainNumber, current->seatsAvailable);
            } else {
                printf("No seats available on %s (Train No: %d).\n", current-
>trainName, trainNumber);
            }
```

```
            return;
        }
        current = current->next;
    }
    printf("Train number %d not found.\n", trainNumber);
}
void displayTrains(struct Train* head) {
    struct Train* current = head;
    if (current == NULL) {
        printf("No trains available.\n");
        return;
    }
    while (current != NULL) {
        printf("Train No: %d, Name: %s, Seats Available: %d\n", current-
>trainNumber, current->trainName, current->seatsAvailable);
        current = current->next;
    }
}
void freeTrains(struct Train* head) {
    struct Train* current = head;
    struct Train* next;
    while (current != NULL) {
        next = current->next;
        free(current);
```

```
        current = next;
    }
}
```

OUTPUT:

main.c

```c
//SHEIKH WASIMUDDIN 23071044 60 CSEIOT//
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Train {
    int trainNumber;
    char trainName[50];
    int seatsAvailable;
    struct Train* next;
};

struct Train* createTrain(int trainNumber, const char*
void addTrain(struct Train** head, int trainNumber, con
void bookTicket(struct Train* head, int trainNumber);
void displayTrains(struct Train* head);
void freeTrains(struct Train* head);

int main() {
    struct Train* head = NULL;
    int choice, trainNumber, seatsAvailable;
    char trainName[50];

    do {

        printf("\nRailway Reservation System\n");
        printf("1. Add Train\n");
        printf("2. Book Ticket\n");
        printf("3. Display Available Trains\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

```

```
Railway Reservation System
1. Add Train
2. Book Ticket
3. Display Available Trains
4. Exit
Enter your choice: 1
Enter Train Number: 12833
Enter Train Name: HOWRAH_SF_EXPRESS
Enter Seats Available: 250
Train added successfully!

Railway Reservation System
1. Add Train
2. Book Ticket
3. Display Available Trains
4. Exit
Enter your choice: 1
Enter Train Number: 20826
Enter Train Name: BSP_VANDE_BHARAT
Enter Seats Available: 300
Train added successfully!

Railway Reservation System
1. Add Train
2. Book Ticket
3. Display Available Trains
4. Exit
Enter your choice: 3
Available Trains:
Train No: 20826, Name: BSP_VANDE_BHARAT, Seats Available: 300
Train No: 12833, Name: HOWRAH_SF_EXPRESS, Seats Available: 250
```

main.c

```c
//SHEIKH WASIMUDDIN 23071044 60 CSEIOT//
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Train {
    int trainNumber;
    char trainName[50];
    int seatsAvailable;
    struct Train* next;
};

struct Train* createTrain(int trainNumber, const char*
void addTrain(struct Train** head, int trainNumber, con
void bookTicket(struct Train* head, int trainNumber);
void displayTrains(struct Train* head);
void freeTrains(struct Train* head);

int main() {
    struct Train* head = NULL;
    int choice, trainNumber, seatsAvailable;
    char trainName[50];

    do {

        printf("\nRailway Reservation System\n");
        printf("1. Add Train\n");
        printf("2. Book Ticket\n");
        printf("3. Display Available Trains\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
```

```
Railway Reservation System
1. Add Train
2. Book Ticket
3. Display Available Trains
4. Exit
Enter your choice: 2
Enter Train Number to book: 12833
Ticket booked on HOWRAH_SF_EXPRESS (Train No: 12833). Seats remaining: 249

Railway Reservation System
1. Add Train
2. Book Ticket
3. Display Available Trains
4. Exit
Enter your choice: 2
Enter Train Number to book: 12833
Ticket booked on HOWRAH_SF_EXPRESS (Train No: 12833). Seats remaining: 248

Railway Reservation System
1. Add Train
2. Book Ticket
3. Display Available Trains
4. Exit
Enter your choice: 3
Available Trains:
Train No: 20826, Name: BSP_VANDE_BHARAT, Seats Available: 300
Train No: 12833, Name: HOWRAH_SF_EXPRESS, Seats Available: 248

Railway Reservation System
1. Add Train
2. Book Ticket
3. Display Available Trains
4. Exit
Enter your choice: 4
Exiting the system.
```

CONCLUSION: This practical was performed successfully.